



“An App That Truly Works for Us”: Co-Designing a Mobile Health Intervention to Support the Transition to Adulthood for the Latino Autism Community

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Abstract

Background: Autistic Latino young adults face significant disparities and a service cliff during their transition to adulthood. While mobile health (mHealth) applications offer a scalable solution, few evidence-based, culturally adapted programs have been successfully transitioned to mHealth platforms for this population. This study aimed to conduct a multi-stakeholder needs assessment to guide the development of a culturally adapted mHealth app designed to accompany the *¡Iniciando!!/Launching!* treatment program to support young adults transitioning to adulthood. **Methods:** This study used a multi-phase, mixed-methods design. An initial needs assessment survey was administered to three key stakeholder groups: autistic young adults ($N=27$), their Spanish-speaking parents ($N=26$), and program clinicians ($N=9$). An iterative refinement process followed, which included qualitative consultations with a Community Advisory Board (CAB) of autistic self-advocates and a User Experience/User Interface (UX/UI) team, a focus group with clinicians ($N=8$), and two targeted follow-up surveys to define specific app features. **Results:** A strong consensus emerged on the need for features supporting executive functioning (e.g., reminders, task management) and practical transition-to-adulthood goals (e.g., employment and education hubs). The iterative design phase translated abstract concepts, like the highly-rated “social games,” into concrete, user-informed specifications, including a scaffolded learning workflow and a gamification-based token economy. Young adults emphasized the importance of a modern user experience and expressed skepticism toward conversational AI, while parents and clinicians stressed that the app must supplement, not replace, human interaction. **Conclusion:** This iterative, community-based approach provides a clear, stakeholder-driven blueprint for creating a highly desired mHealth intervention. By directly integrating the perspectives of end-users, this process ensures the final product is tailored to the specific needs of autistic Latinos, with the potential to reduce health disparities.

Keywords Autism spectrum disorder · MHealth · Latino health · Transition to adulthood · Community-based participatory research

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Autism spectrum disorder (ASD) is a neurodevelopmental condition marked by differences in socialization and repetitive patterns of behaviors/interests (American Psychiatric Association, 2022). ASD affects approximately 3% of children in the United States (US) (Maenner et al., 2023), with a 30% increase in diagnosed cases in US Latinos between 2018 and 2020 (Maenner et al., 2023). Latinos account for 53% of the recent US population growth (Jens Manuel Krogstad et al., 2022). Despite the availability of evidence-based mobile health (mHealth) interventions for Latinos for other health needs (Oswald et al., 2022; Sanchez-Flack et

al., 2021), there are no mHealth apps of culturally adapted programs for young Latino adults with ASD (Arnett, 2003).

Latinos with ASD experience significant disparities in access to autism self-management services across their lifespan due in part to being a hidden population, a group often overlooked in service delivery and research due to geographic isolation, systemic socio-economic barriers, and a lack of culturally tailored outreach (Witherspoon et al., 2020), language barriers, and living in areas with shortages of mental health professionals (Magaña et al., 2013; Zuckerman et al., 2017). One hundred thousand US Latino adolescents with ASD who transition to adulthood each year lose the few available services that they received during their school-age years (e.g., therapy) (Shattuck et al., 2012; Taylor & Seltzer, 2011), and thus face a ‘service cliff’ (Agapie et al., 2022; Arnett, 2003; Dietz et al., 2020; Liu et al., 2023; van Schalkwyk & Volkmar, 2017), while confronting many adult responsibilities for the first time (e.g., daily living skills) (Arnett, 2003; Joyner et al., 2002). This transition occurs in the context of long-standing difficulties with social cognition (Velikonja et al., 2019) and executive functioning (Brady et al., 2017), resulting in mental health distress (e.g., anxiety, depression) (Friedman et al., 2013; Kenworthy et al., 2010; Matthews et al., 2015; McMorris et al., 2019; Meyer et al., 2006; Newman et al., 2011; Rong et al., 2021; Seltzer et al., 2004; Volden, 2004) along with stress associated with marginalized status for parents (Nomaguchi & Milkie, 2020; Zablotsky et al., 2013).

Mobile health (mHealth) applications, defined here as mobile technologies used for the delivery of healthcare services or health-related information, promise to overcome several barriers to care (e.g., geographic, language) confronting Spanish-speaking Latinos with ASD and their parents (Liu et al., 2023). Unlike simple telehealth platforms (video conferencing) or static informational apps, interactive mHealth interventions can deliver therapeutic content, track progress, and provide just-in-time adaptive interventions. Evidence-based interventions adapted to mHealth are particularly useful for Latino young adults with ASD because they: (1) provide support to those in need when and where they need it; (2) amplify resources by providing low cost, scalable interventions; (3) overcome geographic barriers to care; and (4) provide a portable social environment preferred by autistic adults (Hassrick et al., 2021; Valencia et al., 2019). Furthermore, mHealth apps consistently improve therapy homework compliance, defined as the completion of the homework handout provided after every group therapy session, which gives participants the opportunity to practice skills learned in group in their individual lives, which is associated with improved therapeutic outcomes such as improved social functioning, executive

functioning, and adaptive functioning (Agapie et al., 2022; Bunnell et al., 2021; Koh et al., 2022).

While historical concerns regarding a ‘digital divide’ (the gap in access to and knowledge of modern information and communications technology) persist, recent data suggest a shift in technological engagement among Latinos. Latino youth report being ‘almost constantly’ online at nearly double the rate of their White peers (52% vs. 27%; Faverio, 2025). Furthermore, smartphone ownership among Hispanic adults remains exceptionally high at 95% (Flores, 2025), positioning mHealth apps as a primary gateway for service delivery. Culturally adapted and translated interventions are consistently more effective than non-adapted/translated treatments (Griner & Smith, 2006). Such a framework needs to be considered when developing mHealth apps. This is a critical gap, as research regarding general technology use among autistic individuals highlights a significant preference for visual stimuli (Cardon, 2016). Additionally, while broader mHealth literature identifies a growing theoretical potential for Artificial Intelligence (AI) to assist with personalizing content and automating progress tracking (Dhinakaran et al., 2022), there is a lack of research on how these technologies are perceived by specific clinical and cultural groups. This study addresses this gap by exploring stakeholder perspectives on the potential integration of these emerging technologies into culturally adapted interventions, rather than assuming their universal benefit.

The role of parents is uniquely important in this cultural context. For Latinos, the transition to adulthood often occurs within a framework of familismo, emphasizing interdependence rather than complete independence. Furthermore, autistic young adults often need continual parent support longer than their neurotypical peers as they enter adulthood. Therefore, it is essential to include parents, as they are spending the most time with the young adults and supporting them most. Parents of autistic young adults often experience stress associated with marginalized status (Nomaguchi & Milkie, 2020; Zablotsky et al., 2013).

The *¡Iniciando! la Aduldez/ Launching!* to adulthood (Pagán et al. 2024a) therapy program is centered on the transition to adult period and its impacts on both the young adult with ASD and their parent(s). *¡Iniciando!* follows the Pediatric Self-Management model (Modi et al., 2012) by focusing on increasing self-management behaviors in three core areas (Oswald et al., 2018): Co-occurring mental health problems, executive functioning, and social cognition. For Spanish-speaking parents, *¡Iniciando!* focuses on psychoeducation on autism, building skills, and managing stress. Consequently, culturally adapted mHealth apps must not only serve the young adult but also provide services to their parents, who have a support group of their own in the associated *¡Iniciando!* therapy program, giving them an avenue to

support their child's progress (Crosnoe & Cavanagh, 2010) during this high-risk transition (Arredondo et al., 2014; Bishop-Fitzpatrick & Kind, 2017; Magaña et al., 2013). We completed a Phase I clinical trial (NCT06107972, registered on 10/22/2023) with young adult Latinos with ASD and their parents (Pagán et al., 2025). The Spanish version of this therapy program includes 2 in-person (combined parents and young adults) and 10 separate (young adults and parents) virtual meetings over 12 weeks (i.e., hybrid). Young adults showed significant post-treatment improvements on measures of social responsiveness (SRS-2), adaptive behavior (ABAS-3), executive functioning (BRIEF-A), quality of life (WHOQOL-BREF), transition readiness (TRS-A), coping self-efficacy (CSES), and emotion regulation (DERS-SF). Parents reported significant improvements in their perceptions of their child's social responsiveness, adaptive behavior, and executive functioning, as well as increased transition readiness ratings. Notably, parents evinced a significant decrease in acculturative stress (RASI) and depressive symptoms (PHQ-9); however, parental baseline anxiety remained unchanged. Despite these promising outcomes, homework completion rates were poor for young adults (less than 50%). Thus, new strategies are needed to improve engagement. Thus, we decided to develop an mHealth app as a new strategy to improve homework compliance and engagement with the hybrid format.

Many existing mHealth apps are one-size-fits-all (Leask et al., 2019), often lacking the end-user input necessary to maximize their impact (Blandford, 2019). To create an effective and sustainable intervention (An et al., 2023), adaptations of therapy programs into mHealth apps should follow an established framework that centers the user's perspective (Jung & Berthon, 2009). Thus, this study employed a community-based participatory research (CBPR) approach (Wallerstein, 2017) and followed the Integrate, Design, Assess, and Share (IDEAS) framework (Mummah et al., 2016). The aim of this study was to conduct a comprehensive, stakeholder-driven needs assessment to inform development of the *¡Iniciando!* mHealth app, ensuring it is culturally tailored, accessible, and effective in addressing the unique challenges faced by autistic Latino young adults and their Spanish-speaking parents.

Methods

Study Design

This study used a multi-phase, cross-sectional, mixed-methods design framed within the Integrate, Design, Assess, and Share (IDEAS) framework to develop a

technology-enabled health intervention (Mummah et al., 2016). The project followed an iterative process to ensure the *¡Iniciando!* mHealth app was co-designed with its end-users, with feedback gathered continuously across three key groups: the primary stakeholders, a Community Advisory Board (CAB), and a User Experience/User Interface (UX/UI) development team (See Fig. 1 for a flowchart of the study design). All procedures were approved by the Institutional Review Board at the University of Texas Health Science Center at Houston. Former participants from the *¡Iniciando!* Phase I clinical trial with Latinos with ASD, their parents, and coaches (used to describe the role of clinicians working with these clients as non-directive and supportive) were contacted (Pagán et al., 2025). Participants were recruited from a larger clinical trial (Pagán et al., 2025). The current needs assessment focused on a sub-sample of these participants who opted into the design phase, resulting in a final sample of 27 young adults, 26 parents, and 9 coaches. For the young adult group included in this study, the sample was predominantly male (75%). The mean age for the young adult group was 20.53 years ($SD=2.15$). In terms of identity, 100% of the sample identified as Hispanic and White. Regarding educational attainment, the majority were high school graduates (82%), while 13% had completed some high school and 5% held an undergraduate degree. Clinical characteristics measured by the Social Responsiveness Scale Second Edition (SRS-2) showed a mean total score of 74.57 ($SD=16.21$), indicating mild to moderate impairments in reciprocal social behavior. Subscale scores for the young adults included Social Awareness (67.15), Social Cognition (73.47), Social Communication (72.74), Social Motivation (70.51), and Restricted Interests and Repetitive Behavior (74.98). The Community Advisory Board (CAB) members were engaged as expert consultants rather than traditional study participants. An invitation to participate containing a link to the electronic survey was distributed to eligible individuals via email and text message.

Participants and Recruitment

Former participants from the *¡Iniciando!* Phase I clinical trial were contacted for this study. The three distinct stakeholder groups included:

Young Adults Individuals aged 18–25 who identified as Latino and had a confirmed diagnosis of ASD. Participants were verbal and possessed the cognitive and

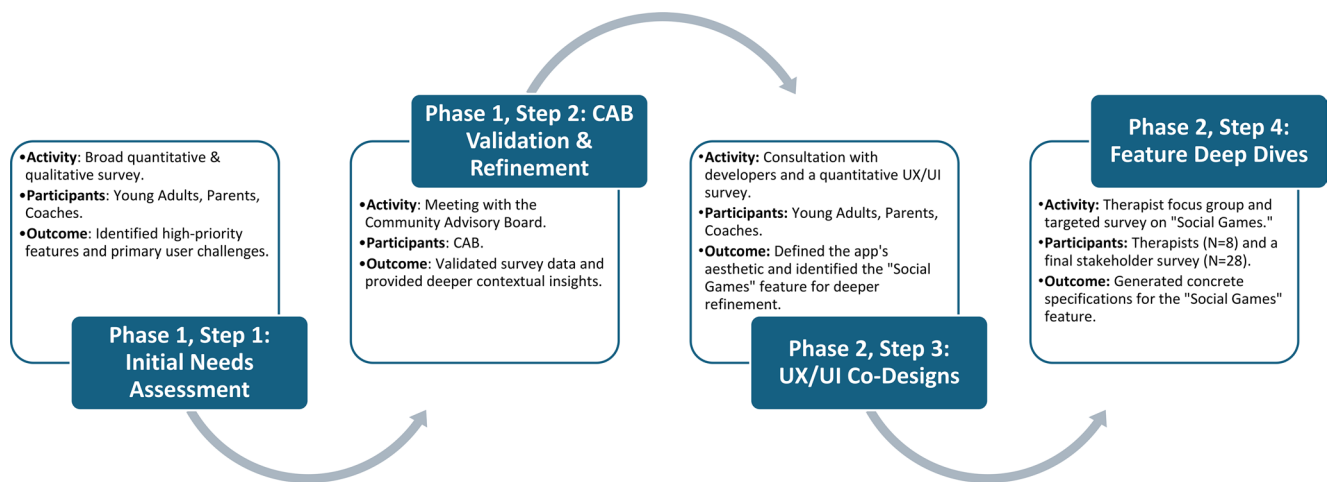


Fig. 1 Visual representation of iterative phase 4 of the IDEAS framework

reading abilities required to complete self-report surveys and participate in group therapy.

Parents Spanish-speaking or English-speaking parents or primary caregivers of the participating young adults.

Coaches Coaches and promotores de salud (community health workers) who delivered the group and individual sessions for the *¡Iniciando!* program. These clinicians are predoctoral psychology interns or master's level psychology graduate students trained under our program's coaching model. The intervention is delivered in a shared partnership model (i.e., working with the client to tackle the problems together) rather than as a typical counselor-client relationship where the counselor finds the solution for the client and tells the client what to change.

Data Collection and Iterative Design Process

The data collection process involved a sequence of three surveys and multiple consultations to move from a broad needs assessment to the specific design of app features.

Phase 1 consisted of a comprehensive, bilingual mixed-methods electronic needs assessment survey developed in Qualtrics and tailored for each stakeholder group with both quantitative questions and open-ended questions. For young adults, the survey included questions about their program experience, barriers to homework completion, and quantitative ratings for over 30 potential app features. The parent version, provided in Spanish, focused on the parent perspective, while the clinician version assessed the app's utility as a professional tool. To fulfill the qualitative aim of this phase, the surveys included open-ended questions designed to elicit deeper insights, such as asking stakeholders to identify

the single most important feature for the app or the biggest challenges to skill generalization. Qualitative data from these surveys consisted of the written text provided by participants in the open-ended fields. The full set of survey items and qualitative prompts for this phase is available in Section S1 of the Supplementary Materials.

Phase 2 involved a multi-step co-design and refinement process. Findings from Phase 1 were first presented to the Community Advisory Board (CAB), which consisted of five autistic self-advocates. This meeting followed a semi-structured interview guide, provided in the Supplementary Materials, which was informed by Phase 1 results to target areas of stakeholder discrepancy, such as the varied perceptions of artificial intelligence. Following CAB validation, findings were presented to the UX/UI development team to assess technical feasibility. These interdisciplinary consultations prompted a second follow-up survey regarding aesthetic style, onboarding methods, and dashboard functionality, with full items listed in Section S2 of the Supplementary Materials. Finally, to define the highly-rated but abstract social games feature, the research team conducted a focus group with clinicians ($N=8$) and a final targeted survey with all stakeholders to specify content and game mechanics. The questions for this focus group were set in advance via a semi-structured guide informed by both behavioral theory and the technical requirements of the development team. The specific specifications for these social games and the associated survey items are detailed in Section S3 of the Supplementary Materials. All qualitative sessions, including the CAB meetings and the clinician focus group, were recorded and transcribed verbatim to ensure an accurate record for analysis.

Data Analysis

Quantitative data from the Likert-scale and multiple-choice questions were analyzed using descriptive statistics. Mean scores, medians, and frequency distributions were calculated for each feature to identify priorities and areas of consensus within and across the three stakeholder groups. Qualitative data from the open-ended questions were analyzed using a thematic analysis approach (Braun & Clarke, 2006). The research team (AFP, MO, GG) reviewed all text responses to identify recurring concepts and patterns. Because the parent survey was administered in Spanish, qualitative responses were translated into English for the purpose of analysis and reporting. Translations were performed by a bilingual member of the research team (AFP and MO) and cross-verified by the team to ensure the original meaning and cultural nuances were preserved. Any disagreements in the coding or theme development were resolved through weekly consensus meetings to ensure inter-rater consistency and thematic depth. To ensure the rigor of the analysis, all members of the coding team possess formal graduate-level training in qualitative research methodologies and have extensive professional backgrounds in neurodiversity-affirming research practices. Through this process, initial codes were generated and subsequently grouped into broader, overarching themes that captured the core priorities, concerns, and innovative suggestions of each stakeholder group. This mixed-methods approach allowed for a robust and nuanced understanding of stakeholder needs to directly inform the development of the ¡Iniciando! mHealth app.

Table 1 Top-rated features from phase 1 survey

Stakeholder group	Rank	Feature	Mean rating (out of 10)
Coaches/Therapists (N=9)	1	"Social Games or interactive scenarios	9.86
	2	Interactive/collaborative homework	9.29
	3	Employment Hub	9.22
Parents (N=26)	1	"Tips for Parents" section	9.70
	2	Notifications on child's goal completion	9.65
	3	Work on a shared goal with child	9.65
Young adults (N=27)	1	Employment Hub	9.08
	2	College and School Hub	8.96
	3	Break down goals into smaller steps	8.79

Results

The final sample for Phase 1 included 9 coaches, 26 parents, and 27 young adults.

Phase 1: Needs Assessment

Young Adults (N = 27)

The young adult stakeholders provided specific and practical feedback, emphasizing the need for a tool that supports self-management, addresses executive functioning challenges, and provides a modern, engaging user experience. They rated features highest that offered practical, real-world support for transition goals, such as the Employment Hub (M=9.08) and the College & School Hub (M=8.96). Features designed to support executive functioning, like the ability to break down large goals into smaller to-do items (M=8.79), were also highly valued. In contrast, features involving AI-driven conversation or goal-setting received the lowest scores (M=5.00–5.92.00.92), suggesting a skepticism toward this type of support compared to more direct, informational tools (Table 1).

Thematic analysis of qualitative feedback from young adults provided a clear mandate for the app's core purpose and design (Table 2). When identifying the single most important feature, seven participants (26%) described tools related to executive functioning, such as a "checklist for assignments" young adult 5 (YA 5) or "reminders" (YA 14). User experience was a prominent theme for five young adults (19%), with feedback emphasizing that the app must feel "smooth to use" (YA 2) and avoid a "clunky" (YA 10) or "outdated" (YA 21) interface. Regarding barriers, six young adults (22%) identified the social demands of group sessions as challenging, noting that high social demands were difficult when "showing up to them with a low social battery" (YA 8). This group also provided a range of novel feature suggestions through open-ended responses, including a Yelp-style rating system for sensory environments (11%), a routine coach with daily task checklists (7%), a tone/phrase helper to suggest social scripts (4%), and customizable avatars for self-expression (4%).

Parents (N = 26)

Parents were overwhelmingly supportive, seeking tools that empower them to support their child's development while finding resources and community for themselves. They rated nearly every proposed feature with exceptionally high scores, indicating a strong perceived need for the app. Features that empowered them to actively support their child were rated highest, including a "Tips for Parents" section

Table 2 Summary of qualitative themes from phase 1

Theme	Young adults & CAB	Parents	Coaches/ Therapists
Frequently Cited Need	Tools for Executive Functioning (reminders, checklists) and a modern, non-"clunky" user experience.	A centralized hub for information and resources on transition and government services.	Overcoming young adults' Executive Functioning deficits (forgetting homework, information overload).
Frequently Cited Concern	Social demands of group sessions causing burnout; skepticism toward "customer support-like" AI.	App supplementing, not replacing, the human connection of therapy.	User privacy in community features and the app devaluing the human element of therapy.
Key Recommendation	A single, unified group chat to avoid overwhelm; notifications should be concise and direct.	Ensure the app is bilingual, reliable, and accessible for all tech-literacy levels.	Create tools for applied practice (like social games) to help generalize skills between sessions.

($M=9.70$), notifications for when their child completes a goal ($M=9.65$), and a feature to work on a shared goal together ($M=9.65$). The ability to see their child's progress ($M=9.61$) and access information on navigating adult services ($M=9.65$) were also rated as extremely helpful.

The most dominant qualitative theme (Table 2) was a concern about the app replacing human interaction, mentioned by seven parents (27%). One parent stated, "La respuesta humana no se debe abandonar, esa es insustituible" (The human response should not be abandoned, it is irreplaceable; Parent 12). Another worried about losing the opportunity for young adults to "socializar personalmente" (socialize in person; Parent 3). When asked about the app's most important function, eight parents (31%) prioritized a reliable resource hub, describing the need for a "biblioteca clara muy ordenada de los recursos que organizaciones y gobierno ofrece" (a clear, very organized library of resources; Parent 18). Five parents (19%) emphasized progress tracking as a key function, requesting an "opcion de hacer seguimiento a tareas y actividades asignadas" (option to follow up on assigned tasks; Parent 5).

Coaches ($N=9$)

The clinical team provided a professional perspective, focusing on the app's utility as a tool to enhance therapy, monitor progress, and generalize skills. Coaches showed strong support for features that provided actionable insights and tools for interactive therapy, with the highest ratings given to features related to practical skill application and progress monitoring. The most valued feature was Social games or interactive scenarios ($M=9.86$), followed by an interactive/collaborative homework feature ($M=9.29$), a comprehensive employment hub ($M=9.22$), and a secure clinician portal to review progress ($M=9.14$). Conversely, features less central to the clinical workflow, such as a values card sort tool ($M=7.67$) and an activity planning feature ($M=7.00$), were rated as less helpful.

When asked about the primary challenge they face in a free-response survey question, all nine coaches (100%) identified issues related to young adults' executive functioning, which directly impacts homework completion and session engagement. Thematic analysis of these qualitative responses revealed that Seven of the nine coaches (78%) noted that young adults forget homework or lose track of materials. One coach detailed this, stating, "YA gets overwhelmed with the amount of emails they receive" (Coach 4), while another emphasized that "the biggest challenge I have is motivation to do the homework in combination with executive function difficulties" (Coach 2). In response to open-ended questions about integration concerns, four coaches (44%) expressed concerns about user privacy, three (33%) cited technical risks, and three (33%) were cautious about the app devaluing the human element of therapy. As one coach explained, "I'm cautious about any tool that might substitute or devalue face-to-face human interaction" (Coach 7). When identifying the most impactful features, four coaches (44%) named social games. Other novel suggestions included a "self-care" hub and a centralized calendar.

Phase 2: Co-Designing the App's Look, Feel, and Workflow

Community Advisory Board Feedback

The initial findings were presented to the Community Advisory Board (CAB) of five autistic self-advocates for validation and deeper insights. The CAB validated that executive functioning was a significant obstacle to homework completion and recommended that in-app notifications be concise and direct, suggesting calendar invitations as a helpful alternative. They proposed gamification elements, such as unlocking assignments sequentially, and suggested

integrating assignments as interactive modules rather than as external documents.

Reflecting the survey data, the CAB expressed nuanced quality concerns about AI, cautioning against “generic, customer support-like answers” (CAB 1) and noting its limited utility for simplistic tasks. They provided critical feedback on social features, strongly recommending a single, unified group chat to avoid overwhelm and foster a stronger sense of

community. The board also offered key insights into mental health features, suggesting that practicing active tools and strategies would be more effective than traditional journaling. They strongly affirmed the high value of the Transition to Adulthood Hubs, explaining that these features support self-direction, which is highly motivating. Finally, the CAB provided actionable recommendations for implementation and safety, including a mandatory user agreement, in-chat reminders of community rules, and entry safeguards to protect user privacy. One member specified, “Have a warning message prior to entering the chat... Don’t share address, personal info” (CAB 5).

Following the initial needs assessment and CAB consultations, the research team engaged in an iterative design process with stakeholders and the UX/UI development team. This involved follow-up surveys to translate broad preferences into specific design features, confirming the feasibility of an adaptable web-based application with features like a permission-based parent dashboard. Participants in this phase had previously participated in the English *Launching!* or Spanish *¡Iniciando!* therapy program and were a part of Phase 1.

Design Preferences

The CAB provided specific feedback on the app’s aesthetic and branding, reaching a consensus for a simple, minimalist, and modern design with customizable color schemes. They issued a clear mandate to not include puzzle pieces in the app’s design, citing the symbol’s controversial history and the stigma it carries for many in the autistic community. Board members explained that the puzzle piece is frequently associated with the medical model of disability and implies autism is a “mystery to be solved” (CAB 2) rather than a natural variation of human neurofunction. To ensure the app felt inclusive and neurodiversity-affirming, the board recommended using the program’s existing rocket logo as a more representative visual. The also suggested drawing inspiration from apps like Duolingo, Blackboard, and My Study Life. They envisioned the app as a one-stop shop that uses clear word prompts for navigation and features a homepage dashboard showing goal and homework status.

A follow-up survey with 28 young adults, 26 parents, and 9 coaches quantified these preferences (Table 3). Young adults confirmed a preference for a “Calm and minimalist” (57%) or “Professional and clean” (43%) style and an “Optional ‘Help’ or ‘Tutorial’ section” (64%) for onboarding. A strong majority (75%) preferred having access to all lessons at once, reinforcing the theme of autonomy, while 68% reported they would not use an AI feature for conversation practice. Parents were nearly evenly split between

Table 3 Stakeholder preferences from the UX/UI Co-design survey

Design category	Young Adults (N=28)	Parents (N=26)	Coaches & Therapists (N=9)
App aesthetic	*Calm and Minimalist" style preferred (57%).	N/A	N/A
Onboarding	Optional "Help/ Tutorial" section preferred (64%).	N/A	N/A
Content access	Strong preference for accessing all lessons at once (75%).	Evenly split between a "Guided Path" (54%) and "Open Access" (62%).	N/A
AI for Conversation	High skepticism, with 68% stating they would not use it.	N/A	N/A
Social features	Favored one large group chat (43%) with features like polls (39%).	N/A	N/A
Communication	N/A	Push notifications were the top choice (73%).	A secure text messaging feature was deemed sufficient.
Clinician dashboard	N/A	N/A	Must provide an at-a-glance view of goals, homework status, and attendance.
Key metrics to track	N/A	N/A	Module completion status (89%) and goal progress (78%).
Homework review	N/A	N/A	A clear consensus for seeing both completion status and the user's actual answers (89%).

favoring a guided path (54%) and open access (62%) for their learning modules and overwhelmingly preferred push notifications (73%) for communication. Coaches wanted a dashboard with at-a-glance information on goal progress and homework completion, and 89% wanted to track module completion status and review both the completion status and the user's actual answers.

Defining the Social Games Feature

To define social games, a focus group with 8 coaches was conducted. Key themes included designing features around making friends and a scaffolded workflow. Therapists recommended a token economy and “Duolingo-style mechanics” (Coach 8). A final survey with 5 coaches, 11 parents, and 12 young adults specified the final design (Table 4). Coaches confirmed the feature should target cognitive flexibility and perspective-taking (100%). Young adults identified “small talk” (YA 28) and “handling disagreements” (YA 26) as key content and preferred a format combining pre-written responses (83%) with typed entries (50%). Parents stressed the need for “user verification” (Parent 22) and “content safety filters” (Parent 14) to ensure a safe environment. Parents also wanted their children to practice conversations related to independence, such as making a doctor's appointment.

Discussion

The aim of this study was to conduct a comprehensive, stakeholder-driven needs assessment to inform the development of the *¡Iniciando!* mHealth app. By employing an iterative methodology rooted in the initial phases of the IDEAS framework (Mummah et al., 2016), we successfully translated the diverse needs of autistic young adults, their parents, and their clinicians into a cohesive and actionable blueprint for the app. Our findings underscore the value of a continuous feedback loop, which allowed us to move beyond surface-level preferences to co-design specific, highly-tailored features. This iterative process was crucial for validating initial ideas, refining abstract concepts, and ensuring the final design was grounded in the lived experiences of its end-users. Our findings must be contextualized within the unique smartphone dependency of the Latino community. While ownership is high, 28% of Latino adults rely solely on their mobile devices for internet access due to a lack of home broadband (Mcclain & Bishop, 2026). This underscores the necessity of designing the *¡Iniciando!* app to function efficiently on mobile data and within a ‘mobile-first’ framework to avoid excluding families without consistent high-speed home internet.

Table 4 Specifications for the "Social Games" Feature

Feature aspect	Young adult feedback (N=12)	Coach & Therapist feedback (N=5 & N=8)	Parent feedback (N=11)
Content/Scenarios	Small talk, handling disagreements, speaking in groups.	Workplace communication, making friends, assertiveness.	Doctor's appointments, customer service, talking to employers.
Format	RPG-style branching narratives; a mix of pre-written choices (83%) and typed responses (50%).	Duolingo-style mechanics with partial points; both fixed and custom scenarios (100%).	A format that builds confidence, such as fill-in-the-blank or multiple choice.
Workflow	Practice first with a coach or peer (50%) before practicing alone.	Scaffolded approach: modeling -> coach practice -> peer practice.	A process that feels safe and supportive for their child.
Motivation/Gamification			A token economy where users earn tokens for participation, redeemable for avatar customizations or internship opportunities.
Safety & Safeguards	Peer practice should be with known or anonymous users.	User reporting features, clear community norms, and clinical oversight.	User verification, content filters, and a "report" button.

The Primacy of Executive Functioning and Real-World Application

The study's findings powerfully align with well-documented research on the challenges autistic young adults face with executive functioning (EF) (Demetriou et al., 2018; Johnston et al., 2019). These EF vulnerabilities, particularly in planning, initiation, and cognitive flexibility, directly complicate the process of setting and achieving goals (Hodgetts et al., 2018; Johnston et al., 2019). The app addresses EF

deficits by providing immediate rewards and feedback, gamified elements that help bridge the gap between effort and long-term outcomes (St. John et al., 2022). By offering step-by-step guidance, such as chunking complex transition tasks into manageable checklists, the app has the potential to provide the necessary scaffolding for autonomy.

The high ratings young adults gave to the Employment and College Hubs, for instance, reflect a desire for support in pursuing long-term goals, which can be difficult to sustain effort toward without immediate rewards (St. John et al., 2022). Furthermore, the specific feature requests from both young adults and the CAB, such as checklists, reminders, and the ability to break down large goals into smaller steps, are direct, practical solutions to the very planning and initiation challenges identified in prior research. Coaches in our study validated this by identifying EF deficits as the single greatest barrier to their clients' success, reinforcing the urgent need for goal-setting strategies tailored to this population. Given the profound impact that the capacity to autonomously achieve goals has on quality of life (Hill et al., 2011), our findings confirm that an effective mHealth tool for this population must function as a practical EF support system, providing the scaffolding needed to navigate the complex demands of emerging adulthood.

The exceptionally high value placed on the Employment Hub ($M=9.08$) and the College & School Hub ($M=8.96$) by young adults directly addresses the service cliff that autistic youth face upon leaving the structured school system (Shattuck et al., 2012; Dietz et al., 2020). The CAB contextualized this finding, emphasizing that these tools support self-direction for a population that often has to navigate complex transition processes with a significant lack of external support. This underscores the critical need for scalable interventions that can bridge the gap in services and resources confronting autistic Latinos as they enter adulthood (Magaña et al., 2013; Zuckerman et al., 2017). By providing accessible, step-by-step guidance, the app can help mitigate the inequities in education and employment that this population disproportionately experiences (Magaña et al., 2016).

To address the digital divide and specific Latino cultural needs, the app was designed to be fully bilingual and family-centered. While mHealth offers accessibility, we must ensure it does not alienate users with lower digital literacy. The app addresses this by using simple, minimalist interfaces requested by the CAB and providing a Tips for Parents section that acts as a bridge, allowing parents, to engage with the content in a supported manner. This aligns with the cultural value of *familismo*, ensuring the intervention supports the entire family unit.

While stakeholders expressed interest in social features, ethical concerns regarding safety and AI were prominent.

To mitigate risks in community features (e.g., group chats), the *¡Iniciando!* app will incorporate safeguards including end-to-end encryption, role-based access controls, and a clear user agreement. While the original in-person therapy program did not utilize AI, the research team explored its potential inclusion in the app based on broader mHealth trends. Regarding AI, the skepticism from young adults suggests that AI features must be transparent and strictly scoped (e.g., used for role-play scenarios rather than therapeutic advice) to ensure user safety and trust.

Balancing Technology with the Human Element

A consistent concern voiced by both parents and clinicians was that the app might replace, rather than supplement, essential human connection. While some research suggests a preference for AI technologies among autistic individuals (Dhinakaran et al., 2022), our young adult participants rated AI-driven chatbots as the least popular features ($M=5.00-5.92$). The CAB clarified this was not a rejection of AI, but a demand for high-quality, purposeful implementation; generic customer support-like AI was seen as unhelpful, whereas AI for specific tasks like summarizing information or providing in-the-moment suggestions during a practice conversation was viewed favorably. Similarly, while autistic adults are known to prefer portable social environments (Hassrick et al., 2021), our CAB advised against fragmented, interest-based chats, recommending a single, unified group chat to avoid overwhelm and foster a stronger sense of community. The overwhelming enthusiasm from parents highlights the unmet need for family-centered support. Their highest-rated features were those that empowered them to help their children, such as the “Tips for Parents” section ($M=9.70$). This aligns with research calling for interventions that provide parents an avenue to support their young adults' progress (Crosnoe & Cavanagh, 2010). However, the most dominant qualitative theme from parents, mentioned by seven respondents, was a fear that the app would replace essential human interaction. This critical feedback cautions against purely technological solutions and reinforces the value of the *¡Iniciando!* program's hybrid model, where the app serves to supplement, not supplant, the therapeutic relationship with clinicians and the vital connection with peers. The in-app chat and parent portals address the need for family-centered support by creating a verified, safe space for connection, reducing the isolation often felt by Latino families of autistic adults.

Strengths and Limitations

The primary strength of this study is its deep grounding in a community-based participatory research approach, adhering to the IDEAS framework (Mummah et al., 2016) by engaging end-users from the very beginning. By collecting and synthesizing feedback from young adults, parents, clinicians, and an expert self-advocate advisory board, we mitigate the risk of developing a one-size-fits-all application that lacks relevant end-user input (Leask et al., 2019). This process directly addresses recommendations for creating sustainable and impactful mHealth interventions. A primary limitation is that our sample consisted of young adults who were verbal and capable of completing self-report surveys, which limits generalizability to autistic individuals with higher support needs or intellectual disabilities. Additionally, the sample was drawn from a single program, potentially reflecting a pre-existing positive disposition toward the intervention.

Future Directions and Conclusion

The findings provide a stakeholder-driven blueprint for the *¡Iniciando!* mHealth app. By centering the voices of autistic young adults and their families, this project aims to create an accessible, scalable, and culturally congruent tool that can meaningfully reduce health disparities for autistic Latinos transitioning to adulthood. Future research will involve building the app based on these specifications, followed by rigorous usability testing and a feasibility trial to assess engagement, satisfaction, and preliminary outcomes. To ensure that the *¡Iniciando!* app is both usable and accessible for autistic Latino young adults and their Spanish-speaking parents, future studies will implement a multi-method usability evaluation strategy tailored to cultural and neurodiverse needs to evaluate the usability and accessibility of the application. This includes cognitive walkthroughs, which simulate user interactions to identify barriers in task flow and comprehension, particularly relevant for users with executive functioning challenges common in autism, while incorporating culturally specific scenarios such as bilingual navigation and family-centered goal setting (De Leonardis, 2025). Heuristic evaluations, guided by adapted usability principles for neurodiversity, will assess interface design for consistency, error prevention, efficiency, and cognitive inclusivity, ensuring elements like color schemes and iconography resonate with Latino cultural contexts (Maroto-Gómez et al., 2025). We will also use think-aloud protocols, allowing participants to verbalize their thoughts while navigating the app, providing insight into their decision-making, emotional responses, and cultural interpretations

of content to refine evidence-based interventions like self-management strategies (Chandrashekar et al., 2018). Additionally, accessibility audits aligned with the latest Web Content Accessibility Guidelines (WCAG 2.2) will ensure the app accommodates neurodiverse users by minimizing sensory overload, simplifying navigation, supporting cognitive accessibility, and addressing language accessibility for Spanish speakers (W3C, 2023). These methods will be complemented by iterative feedback from the Community Advisory Board, comprising Latino autistic individuals, parents, and cultural experts, to ensure the app remains grounded in the lived experiences of its intended users, reflects inclusive design principles, and incorporates cultural adaptations of evidence-based interventions for improved engagement and outcomes (Magaña et al., 2021).

To ethically implement community features such as group chats and peer forums, the *¡Iniciando!* app will incorporate safeguards including end-to-end encryption, role-based access controls, and a clear user agreement outlining behavioral expectations. These measures are especially important for autistic users, who may be more vulnerable to oversharing or misinterpreting social cues in digital spaces (Alhammad et al., 2024; Hassan et al., 2025). Privacy policies will be written in plain language and aligned with best practices for transparency and user consent in mHealth applications (Largent et al., 2024). These strategies aim to protect sensitive data while fostering a safe and inclusive digital environment. By centering the voices of autistic young adults and their families, this project aims to create an accessible, scalable, and culturally congruent tool that can meaningfully reduce health disparities and improve the quality of life for autistic Latinos transitioning to adulthood.

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Data Availability Data is available upon reasonable request.

Declarations

Ethical Approval All procedures performed were in accordance with the ethical standards of the institutional review boards of the University of Texas Health Science Center Houston and with the 1964 Helsinki declaration and its later amendments.

Informed Consent Informed consent was obtained from all individual participants included in the study.

Clinical Trial Number Not applicable.

Conflict of Interest The authors declare that they have no conflict of interest.

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